Figure 7: Practical Real-World Edge Deployment & In-Line Mitigation Architecture

Kernel eBPF Telemetry Ingestion • Zero-Copy Ring Buffers • INT8 Quantized In-Line Inference • Dynamic iptables Defense

Raspberry Pi 4 / ARM Cortex

Linux eBPF / XDP Ready

LAYER 1: PHYSICAL IOT INFRASTRUCTURE
Field Devices & Adversarial Threat Vectors

Legitimate IoT Nodes
- Continuous telemetry flows
- ThingSpeak cloud updates
- MQTT pub/sub streams
- Smart bulbs & HVAC nodes

Cyber Threat Actors
- Volumetric SYN flooding (Hping)
- Slowloris HTTP starvation
- ARP cache spoofing probes
- Nmap multi-vector scans
- SSH credential brute forcing

Wire Traffic

Attack Ingress

LAYER 2: EMBEDDED EDGE GATEWAY RUNTIME
In-Line Packet Processing & INT8 Inference

eBPF Kernel Ingestion
- Zero-copy ring buffer
- 21 continuous flow metrics
- High-speed kernel hooking
- Sub-microsecond parsing

Raw Stats

RobustScaler Normalizer
- Frozen training medians
- IQR interquartile scaling
- Outlier clamping shielding
- Transductive leakage-free

Normalized

In-Line Neural Engine (INT8 Quantized Edge Gateway)
- Raspberry Pi 4 / Embedded Linux (13,569 parameters)
- INT8 dynamic quantization: Compressed to 48.6 KB (2.69x smaller)
- Single-flow inference latency: 15.2 us (Host) / 731.6 us (Quantized)
- Throughput: >255,000 flows/sec (exceeds Gigabit line-rate)
- Calibrated decision boundary: P(Anomaly) > tau_decision

Alert: Block IP

OTA Updates

LAYER 3: ACTIVE DEFENSE & SOC LOOP
Sub-Second Mitigation & Expert Triage

Automated Mitigation
- Dynamic iptables rule injection
- Immediate MAC / IP quarantine
- TCP RST session teardown
- Zero false interruption target

Forensic Trigger

SOC Human-in-the-Loop
- High-confidence incident triage
- Forensic database storage
- Active learning labeled reservoir
- Periodic cloud SSL retraining
- OTA edge model checkpoint push